REQUIREMENT ANALYSIS

Data Flow Diagram & user stories

Team ID	LTVIP2026TMIDS36743
Project Name	visualization tool for electric vehicle charge and range analysis

Data Flow Diagram & user stories

The EV Charge and Range Analysis System collects real-time data from the vehicle's sensors, charging stations, and external sources like traffic and weather services. It processes this data to estimate the remaining driving range, suggest optimal charging stations, and provide trip planning. Users interact through a mobile or web app where they input preferences and receive notifications, alerts, and visualizations about battery status, charging availability, and route guidance.

EV Charge and Range Analysis Data Flow Diagram

1.User (EV Driver / Researcher)

The primary actor interacting with the system.

Vehicle preferences, trip plans, route choices, personal settings.

Receives battery status, charging station suggestions, route guidance, and alerts.

To plan trips efficiently, monitor EV range, and reduce range anxiety.

2.EV Onboard System / Vehicle Sensors

Internal sensors and onboard diagnostics of the electric vehicle.

None directly; it continuously collects vehicle data.

Sends battery percentage, speed, location, and energy consumption data to the system.

Provides real-time vehicle status for accurate range prediction and analysis.

3.Charging Station Network

External infrastructure providing EV charging services.

User queries for nearby charging stations.

Sends charger availability, type (fast/slow), location, and operational status.

4. Traffic and Weather Data Services

External data sources providing environmental conditions.

System requests for relevant route conditions.

Real-time traffic updates, road conditions, and weather data.

To improve range estimation and suggest optimized routes by considering traffic delays or adverse weather.

5. **EV Charge & Range Analysis System**

Central processing system of the application.

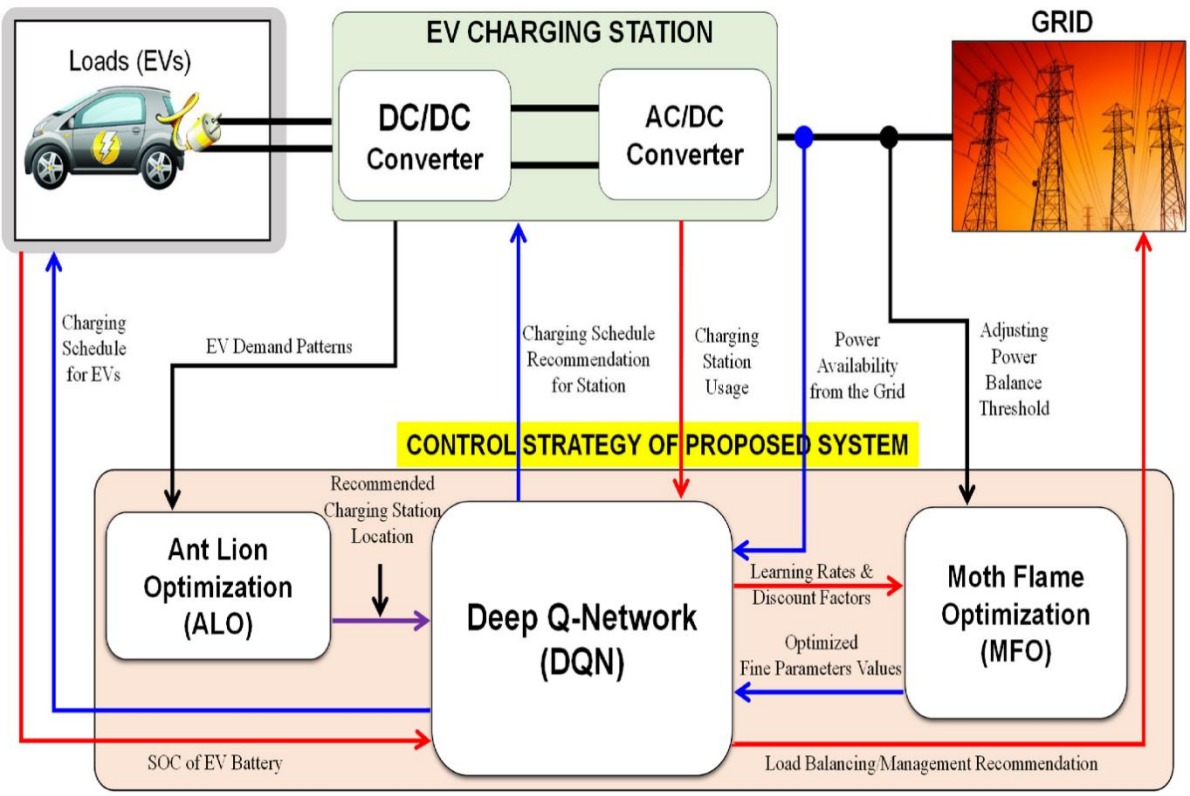
Data from EV sensors, charging stations, traffic/weather services, and user preferences.

6.User Interface (Mobile App / Web Dashboard)

Interface for interaction between the user and the system.

Receives user inputs like preferences, trip plans, and filters.

Displays battery status, range prediction, charging station locations, route guidance, and alerts.


Loads (EVs)
EV CHARGING STATION
DC/DC Converter
AC/DC Converter
GRID
Charging Schedule for EVs
EV Demand Patterns
Charging Schedule Recommendation for Station
Charging Station Usage
Power Availability from the Grid
Adjusting Power Balance Threshold
CONTROL STRATEGY OF PROPOSED SYSTEM
Recommended Charging Station Location
Ant Lion Optimization (ALO)
Deep Q-Network (DQN)
Learning Rates & Discount Factors
Optimized Fine Parameters Values
Moth Flame Optimization (MFO)
SOC of EV Battery
Load Balancing/Management Recommendation